(5) $\lim_{x\to\infty}\left(1+\frac{3}{x}\right)^{x+2}$

(6) $\lim_{x\to 0}\frac{\sin^3\left(\frac{x}{2}\right)}{x^3}$

(7) $\lim_{x\to 0}\frac{\sin\alpha x}{\sin\beta x}$

(8) $\lim_{x\to 0}\frac{\tan 2x}{\sin 5x}$

(9) $\lim_{\alpha\to 0}\frac{\sin(\alpha^n)}{(\sin\alpha)^m}$

(10) $\lim_{x\to 0}\frac{\sin(a+x)-\sin(a-x)}{x}$

(11) $\lim_{x\to 0}\frac{\sqrt{x^2+a^2}-a}{\sqrt{x^2+b^2}-b}$

(12) $\lim_{x\to 0}\frac{2\arcsin x}{3x}$

(13) $\lim_{x\to 0}\frac{1-\cos x}{x^2}$

(14) $\lim_{x\to 0}\frac{\tan 2x}{x}$

(15) $\lim_{x\to 0}\frac{2^x-3^x}{x}$

(16) $\lim_{x\to 0}\frac{3^x-1}{\sqrt{x+1}-1}$

(17) $\lim_{x\to 0}\frac{1-\cos^2 x}{x\sin 2x}$

(18) $\lim_{x\to\infty} x\left[3^{\frac{1}{x}}+1-\cos\left(\frac{1}{x}\right)-e^{\frac{1}{x}}\right]$

(19) $\lim_{x\to\infty}\{x[\log(x+a)-\log(x)]\}$

(20) $\lim_{x\to\pi}\frac{\sin 3x}{\sin 2x}$

(21) $\lim_{x\to\pi}(1+\sin x)^{2\operatorname{cosec} x}$

(22) $\lim_{x\to 0}\frac{\sqrt{2}-\sqrt{1+\cos x}}{\sin^2 x}$

(23) $\lim_{x\to 0}\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{\tan x}$

(24) $\lim_{x\to\infty}\left(\frac{x^2-2x+1}{x^2-4x+2}\right)^x$

(25) $\lim_{x\to 0}\frac{e^x-e^{-x}}{\sin x}$

(26) $\lim_{x\to 0}\frac{e^{ax}-e^{bx}}{x}$

(27) $\lim_{x\to 0}\frac{\sin x(1-\cos x)}{x^3}$

(28) $\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$

9.3 Continuity

One of the chief features in the behaviour of functions is the property known as continuity. It reflects mathematically the general trait of many phenomena observed by us in nature. For instance, we speak of the continuous expansion of a rod on heating, of the continuous growth of an organism, of a continuous flow, or a continuous variation of atmospheric temperature etc.

The idea of continuity of a function stems from the geometric notion of “no breaks in a graph”. In fact, the name itself derives from the Latin continuere, “to hang together”. Nevertheless, to identify continuity with “no breaks in a graph” or “a hanging together” has serious drawbacks, at least from the point of view of applying the concept to the analysis of functions. Accordingly, a premature use of the graph to gain insight into the meaning of continuity is advised against as gravely misleading. However, we will realise later that for functions with interval domains, continuity means essentially that the graph may be traced without lifting the point of the pencil.

The proper and effective way of attitude which allows us to put the concept to work is to correlate continuity with limit. Loosely speaking, to possess the property of continuity will mean “to

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